Ahmedabad University

CSE 516 Probabilistic Graphical Models · Group 5 (G5_TH_PGM)

Deep End-to-end Causal Inference

Milestone 2: Technical Foundation. Auditing the graph-recovery guarantee of DECI (Theorem 1)

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Theoretical AI · M2 due 11 October 2026

M2 focus, questions and contributions

Focus

- Technical implementation and evaluation of the DECI graphical model: representation, learning, inference and baseline-based validation.

Fundamental question

- Under which conditions does the ELBO optimum recover the true causal DAG, and where does the proof stop applying?

Research questions

- Q1 Can Theorem 1 be re-derived, and which step uses which assumption?
- Q2 Mixed type: what fails? Q3 Missing data: which step fails?
- Q4 Do the boundaries show up in graph recovery and ATE/CATE?

Contributions of M2

- Proof reconstruction of Theorem 1 in five steps with an assumption-to-step map.
- Audit: DAG-only likelihood, tightened KL lemma, implicit assumptions A6–A10.
- Boundary propositions for categorical data and amortised imputation.
- Pipeline, KPIs, six falsifiable predictions, first experiments E1 and E4–E5.

Changes from M1

- Scope unchanged. New KPIs: posterior mass on G^{o}, cyclic-sample rate, ordering accuracy.
- Experiments are evidence, not proof; benchmark scores never establish the theorem.

Background, related work and SOTA anchor

Background

- DAG + Markov: p(x)=∏p(x_i|pa); d-separation gives CIs; do(x_T=a) cuts edges into x_T.
- Identifiable: non-linear ANM, linear non-Gaussian. Not: linear Gaussian (only the Markov equivalence class).
- ELBO = log p(X) − KL(q ‖ p(L|X)); latent L = graph (and unobserved entries).

Related work

- PC, GES, LiNGAM, ANM; NOTEARS uses h(G)=tr(e^{G∘G})−D; NOTEARS-MLP/Sob, GraN-DAG for non-linear.
- Geffner Lemma 2: these methods are MLE of a flow with hard acyclicity; DECI adds noise flexibility and a graph posterior.

SOTA anchor: Microsoft Research, ICLR 2025

- Vashishtha et al., Causal Order: pairwise expert queries create cycles; triplet prompts give fewer cycles and more accurate orders.

M2 update

- A DAG sample defines admissible causal orders, so both works share one object.
- New KPI: ordering accuracy of DECI posterior samples (planned for M3).

Relationship

- We make no claim to beat the anchor: it uses expert information, we use data and a likelihood. We audit DECI itself (Geffner et al., 2022).

DECI as a PGM: what each element means

Element	Meaning in DECI	Why this representation
Node x_i	Observed variable (continuous; categorical in extension)	Unit at which interventions act
Edge $j \to i$	x_j is an input of f_i; absent edge forces $\partial f_i/\partial x_j = 0$	A missing edge is a causal and CI claim
f_i	$\zeta_i(\Sigma_j G_{ji}\, \ell_j(x_j))$; residual MLPs, shared weights, node embeddings	Universal approximation, O(D) networks
Noise z_i	Gaussian or 8-bin spline flow, independent across i	Independence gives identifiability
Prior $p(G)$	$\propto \exp(-\lambda_s \lVert G \rVert^2 - \rho h(G)^2 - \alpha h(G))$	Sparse and softly acyclic
Posterior $q(G)$	Independent Bernoulli per edge (ENCO)	Tractable; ignores edge dependence

Assumes causal Markov, minimality and causal sufficiency (no latent confounder). $p_\theta(x \mid G) = \prod p_z(x_i - f_i(x_{pa}))$. One global graph latent is shared by all N samples; this drives the proof and the missing-data contrast.

Learning and inference

Learning (structure + parameters)

- ϕ: graph posterior $q_\phi(G)$. θ: MLPs, embeddings, noise. Learned jointly.
- ELBO = $E_q[\log p(G) \prod p_\theta(x_n \mid G)] + H(q) \le \log p_\theta(x)$; exact marginal sums $2^{(D^2)}$ graphs.
- Gumbel-softmax (τ=0.25), Adam lr 0.01; augmented Lagrangian ρ=1, α=0: if P2<0.65·P1, $\alpha \leftarrow \alpha + \rho P2$ else $\rho \leftarrow 10\rho$.
- Missing: Gaussian $q_\psi(x_u \mid x_o)$. Mixed: p=softmax(f_i).
- Config: MLPs 2×128 residual+LayerNorm, spline 8 bins, λ_s=5, S=100 posterior samples.

Inference

- Queries: q(G); ATE and CATE marginalised over graphs.
- ATE: G~q, z~p_z, cut edges into x_T, fix x_T, simulate topologically; 1000 graphs × 2 samples.
- CATE: joint samples under do(); surrogate g_G with 3000 random Fourier features; 10 graphs × 10 000.
- Why simulation: exact graph inference is exponential and flows have no closed form; fit once, answer many queries. Cost: MC variance.
- Complexity: $O(D^2)$ forward, $O(D^3)$ matrix exponential in h(G).

Proof chain: Theorem 1 in five steps

1 Valid density

- Topological order ⇒ $\partial f/\partial x$ strictly lower-triangular ⇒ $\det(I-\partial f)=1$ ⇒ $p(x|G)=\prod p_z(x_i-f_i)$. DAGs only.

2 Prior finite

- $G\in\{0,1\}^{(D\times D)}$ is finite, so h, $\|G\|^2$, Z are finite. $h\geq 0$, $=0$ iff DAG.

3 Gibbs / MLE

- $E \log p_\theta - E \log p^o = -KL \leq 0$, equal iff $p_\theta=p^o$. A3 attains it, so MLE gives p^o.

4 Identifiability

- A1, A2: only G^o generates p^o (non-linear ANM, non-constant f).

5 KL/N → 0

- N^{-1}ELBO → $\Sigma q(G)E \log p_\theta(x|G)$; maximal only if supp $q=\{G^o\}$, so $q=\delta(G^o)$.

- A1 minimality · A2 structural identifiability · A3 correct specification · A4 causal sufficiency · A5 regularity $E|\log p_\theta|<\infty$
- Map: A3–A5 → step 3 · A1–A2 → step 4 · finite graph set + fixed λ,ρ,α → step 5 · implicit A6–A10 → step 5.

Audit: tightened lemma and hidden assumptions

Lemma 2 (tightened)

- $KL(q\|p) \le \max_G E(G) + \log Z$, $E=\lambda\|G\|^2+\rho h^2+\alpha h$.
- Independent of N and q, so $KL/N \to 0$ uniformly.
- Original needed $w_\phi>0$ and Cauchy–Schwarz.
- Caveat: re-check if ρ, α grow with N (capped 1e13).

Remark 1: cyclic graphs

- Jacobian = 1 only on DAGs. A k-cycle gives $h \ge 1/(k-1)! \ge 1/(D-1)!$.
- D=16: 7.6×10^{-13}. D=64: about 5×10^{-88}. Long cycles are barely penalised at α=1e13.
- Test: cyclic-sample rate (P6).
- Remark 2: closed-walk counting in the original is inexact; finiteness of the graph set suffices.

Implicit assumptions

- A6 uniform LLN over θ
- A7 point-mass q representable (sigmoid edges reach it only in the limit)
- A8 posterior supported on DAGs
- A9 global optimum reached (ELBO non-convex)
- A10 λ, ρ, α fixed in N

Proof boundaries: mixed type and missing data

Mixed type: step 4 fails (Prop. 2)

- Steps 1, 3, 5 survive: Gibbs holds for any reference measure.
- Two binary variables: X1→X2 and X2→X1 each cover every joint, so equal maximum likelihood and no identifiability.
- Expect concentration on the Markov equivalence class, as in non-identifiable lingauss.
- P4: adjacency F1 → 1, orientation stalls at class level.

Missing data: step 5 fails (Prop. 3)

- Under MCAR, gap = $\Sigma_n \mathrm{KL}(q_\psi(x_u \mid x_o) \,\|\, p_\theta(x_u \mid x_o, G^o))$.
- N local latents give $\Theta(N)$; the graph KL is $O(1)$. So $N^{-1} \cdot \text{gap} \not\to 0$ and ELBO optimum need not equal the MLE.
- Non-Gaussian conditionals ⇒ positive gap; θ biased as in VAEs. Zero for linear Gaussian (control).
- MAR/MNAR would mis-specify the objective.

Six testable predictions

ID	Prediction	Exp	KPI	Refuted if
P1	Identifiable ANM: expected SHD falls with n	E1	E_q[SHD] vs n	SHD plateaus well above 0 with converged optimisation
P2	Linear Gaussian: orient. F1 ≈ 0.5, mass on G^o ≈ 1/|MEC|	E5	Orient. F1, mass	Posterior concentrates confidently on one DAG
P3	Gaussian-noise DECI orients worse under non-Gaussian noise	E4	Orient. F1	No gap between DECI-G and DECI-S
P4	Categorical pairs: orientation stalls at class level	E6	Orient. F1, entropy	Orientation recovered reliably
P5	Missingness degrades KPIs smoothly, more if non-Gaussian	E7	F1, ATE RMSE	Abrupt collapse or no degradation
P6	Cyclic hard samples grow with D	E2	Cyclic rate	Rate 0 across D

Implementation pipeline and owners

Pipeline

- Data → preprocess → DECI (Causica) → posterior graph samples → KPIs → ATE/CATE → seeds → plots and tables.
- Reproducible configuration + seed; software versions and commands logged.

Owners and status

- Data generator and ANM: Yashvi Patel, code written and executed.
- DECI-G / DECI-S: Khushi Thakar, code written and executed.
- Baselines NOTEARS-MLP, PC, ICALiNGAM: Masoom Choksi, code written and executed.
- Theory and boundaries: Parisha Kathiria, Vrunda Patel.
- KPI and statistics scripts: all, scaffold.

E1 setup

- ER(16,16), non-linear ANM, spline-warped non-Gaussian noise; $n \in \{250, 500, 1000, 2500, 5000\}$.
- Methods: DECI-S, DECI-G, NOTEARS-MLP, PC, ICALiNGAM; metric: expected SHD, mean ± SD.

Control pilot

- ER, 16 variables, 16 edges, Gaussian noise, 5000 train / 1000 val / 1000 test, seed 42; true graph used only for evaluation.

CSuite (E4–E5)

- lingauss, linexp, nonlingauss, nonlin_simpson; 2000 samples; DECI-G vs DECI-S.

Code: KPIs and experiment loop

```
def shd(Ahat, A):
  diff = Ahat != A
  rev = (Ahat==1)&(A.T==1)&(A==0)
  return int(diff.sum() - rev.sum())

def posterior_kpis(samples, A):
  return dict(
   exp_shd=mean(shd(G, A)),
   adj_f1=mean(adjacency_prf(G, A)[2]),
   orient_f1=mean(orientation_prf(G,A)[2]),
   mass_true=mean(array_equal(G, A)),
   cyclic_rate=mean(not is_dag(G)),
   ordering_acc=mean(ordering_accuracy))

def experiment_sample_size(methods, ns, seeds):
  for seed in seeds:
    A = sample_er_dag(16, 16, rng)
    for n in ns:
      X = simulate_anm(A, n, rng, noise)
      for name, fn in methods.items():
        kpis = posterior_kpis(fn(X, seed), A)
```

What it does

- KPIs are expectations over S posterior samples.
- PC CPDAG: all member DAGs equally likely.
- aggregate: mean, SD, 95% CI (t); paired Wilcoxon if n ≥ 6, else bootstrap.
- Adapters run_deci, run_notears_mlp, run_pc, run_icalingam connect to Causica and gcastle.
- One script per table: python run_e1.py --config configs/e1.yaml.

Experimental setup, baselines, KPIs, protocol

Baselines

- NOTEARS-MLP (primary non-linear), PC, ICALiNGAM; DoWhy linear/DML for ATE in M3.
- All methods get the same data, splits and seeds; unmatched assumptions are noted.

KPIs

- SHD, adjacency/orientation F1, precision, recall, posterior mass and entropy, cyclic-sample rate, ordering accuracy, ELBO/NLL, ATE/CATE RMSE, runtime.

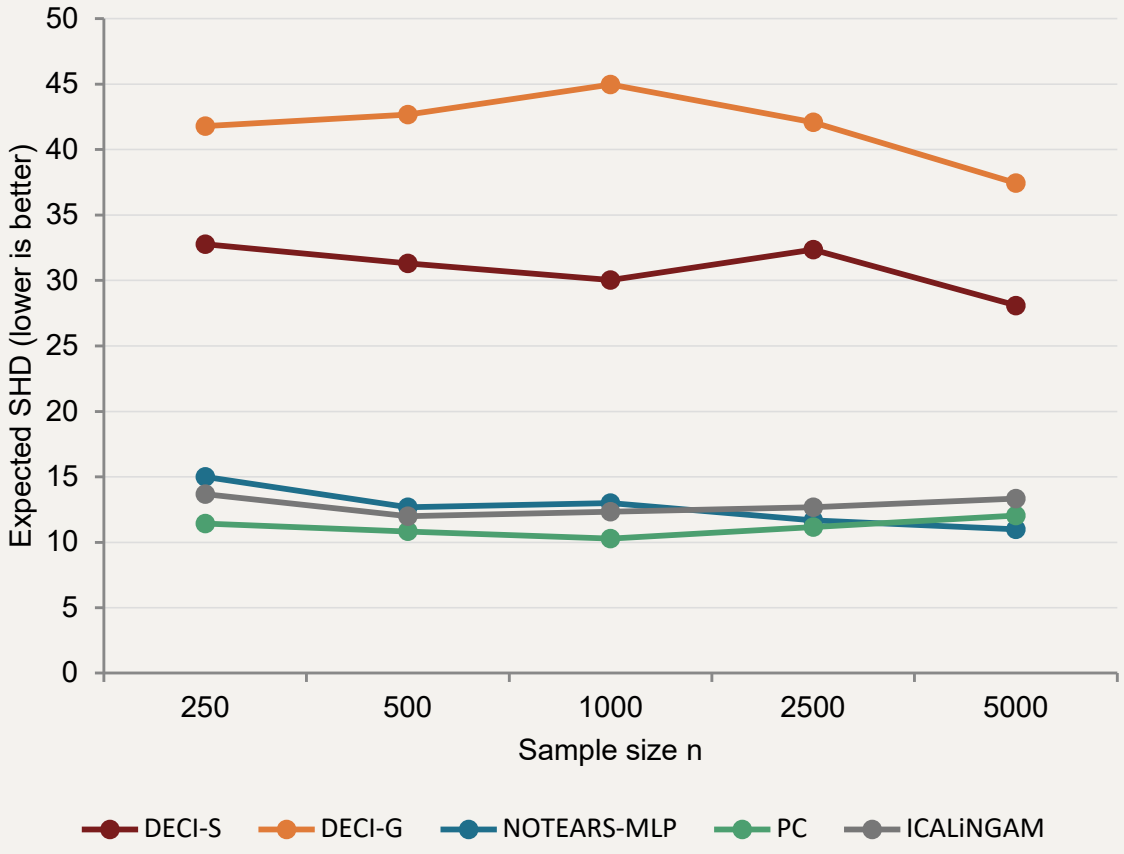
Protocol

- ≥5 seeds per cell (paper: 5 for ER/SF, 20 for CSuite); seeds fixed in config.
- Defaults unless a pilot shows non-convergence; tuning applied symmetrically.
- Runs over budget reported as missing.

Statistics

- Mean ± SD, 95% CI (t or 2000-sample bootstrap); Wilcoxon for ≥6 pairs else effect size + bootstrap CI; slope of log SHD on log n; significance never replaces assumption checks.

E1: sample size vs structural recovery



Reading the chart

- Means ± SD at n=5000: DECI-S 28.09 ± 7.89, DECI-G 37.44 ± 1.12, NOTEARS-MLP 11.00 ± 0.00, PC 12.04 ± 2.77, ICALiNGAM 13.33 ± 2.08.
- DECI trends down but not monotonically.
- PC and NOTEARS-MLP are lower at every n on this 16-edge graph.

E4–E5: CSuite noise model and identifiability

Dataset	Method	Adj. F1	Orient. F1	Mass on G^o	ATE RMSE (ours)	ATE RMSE (published)
lingauss	DECI-G	0.950	0.843	0.750	0.136	0.120
lingauss	DECI-S	0.920	0.840	0.760	0.169	0.149
linexp	DECI-G	1.000	0.013	0.000	0.989	1.029
linexp	DECI-S	1.000	1.000	1.000	0.014	0.022
nonlingauss	DECI-G	1.000	1.000	1.000	0.005	0.042
nonlingauss	DECI-S	1.000	1.000	1.000	0.016	0.043
nonlin_simpson	DECI-G	0.739	0.235	0.000	1.855	1.995
nonlin_simpson	DECI-S	0.859	0.498	0.000	0.889	1.997

linexp: DECI-G finds the skeleton but the wrong direction (orientation F1 0.013); the spline noise model fixes it.
nonlin_simpson stays hard for both: mass on the true graph is 0.

What the results mean

E1 (P1 partially supported)

- What happened: DECI SHD drifts down with n but not monotonically; PC and NOTEARS-MLP are lower throughout.
- Why: finite-sample variance of the Bernoulli posterior, default hyper-parameters, no global optimality (A9), possible extra edges (SHD above 16 edges).
- Lesson: more data alone does not guarantee recovery; the theorem is about the optimum, not what Adam finds.

CSuite (P2, P3)

- P3 supported: linexp orientation F1 0.013 (Gaussian) vs 1.000 (spline), a misspecified noise model.
- Identifiable cases linexp-S and nonlingauss reach mass 1.0.
- P2 as stated is not matched: lingauss mass is about 0.75 and orientation F1 about 0.84, not near 0.5. To investigate: seeds, edge thresholding, sparsity prior.
- nonlin_simpson: Spline beats Gaussian but both fail on orientation and mass.

Evidence vs predictions and experiment status

Prediction	Status	Evidence
P1 sample size	Partially supported	E1: downward trend, noisy; baselines lower
P2 non-identifiable	Not matched yet	lingauss mass 0.75/0.76, orient. F1 0.84
P3 noise model	Supported	linexp: 0.013 vs 1.000 orientation F1
P4 categorical	Not run	E6 planned for M3
P5 missingness	Not run	E7 planned for M3
P6 cyclic samples	Not run	E2 planned for M3

Still to do: log-SHD vs log-n slope with CI, paired Wilcoxon or bootstrap vs NOTEARS-MLP, 95% CIs for Tables 4 and 5, and a seed count consistent across text and captions.

Scope and limitations

Scope

- Audit and reconstruction of the existing DECI guarantee; boundaries for mixed-type and missing data.
- No new universal theorem without a full proof; benchmark scores never prove Theorem 1.
- Mixed/missing experiments are empirical boundary cases, not a theorem extension.

Limitations

- Propositions 2–3 and Remark 1 are our claims, not yet peer-checked; finalised in M3.
- The theorem concerns global optima; experiments cannot separate finite-sample from optimisation effects (A9).
- Mean-field q ignores edge dependence; mass on G^o is a property of q_ϕ.
- Compute may limit D and seeds; missing cells will be reported.
- Baselines differ in assumptions: compare within matched regimes.

Plan, team accountability, reproducibility

Plan

- M3, 1 Nov: E2, E6, E7; full benchmarks (ER/SF, SynTReN, Sachs, IHDP, Twins); ablation; error analysis; statistics; SOTA repositioning with ordering accuracy.
- M4, 22 Nov: complete analysis, KPI plots and tables, limitations, reproducibility, final SOTA position.

Team

- Each member explains the model, factorisation, CIs, learning, inference, KPIs; individual Google Form each milestone.

Reproducibility

- Repo: /theory /src /configs /results /plots /docs.
- One script per table; fixed seeds; Python, PyTorch, Causica commit and gcastle versions copied from the environment log.
- Datasets: CSuite, ER/SF (Geffner et al.), SynTReN, Sachs, IHDP, Twins.
- AI use declared by every member in the individual form.

Key takeaway

- DECI's guarantee is a chain: Gibbs step, identifiability, global-latent KL.
- Categorical data breaks identifiability; amortised imputation breaks ELBO→MLE; the likelihood step survives both.
- First evidence: noise-model misspecification hurts orientation (linexp); on E1, DECI trails PC and NOTEARS-MLP.
- M3 tests the remaining predictions with robustness and statistics.

Thank you